

Question 1

a) The gene map of a plasmid, which has been used for genetic engineering, is shown below in Fig. 1.1. It contains two antibiotic resistance genes. The positions of several restriction endonuclease binding sites are also shown below.

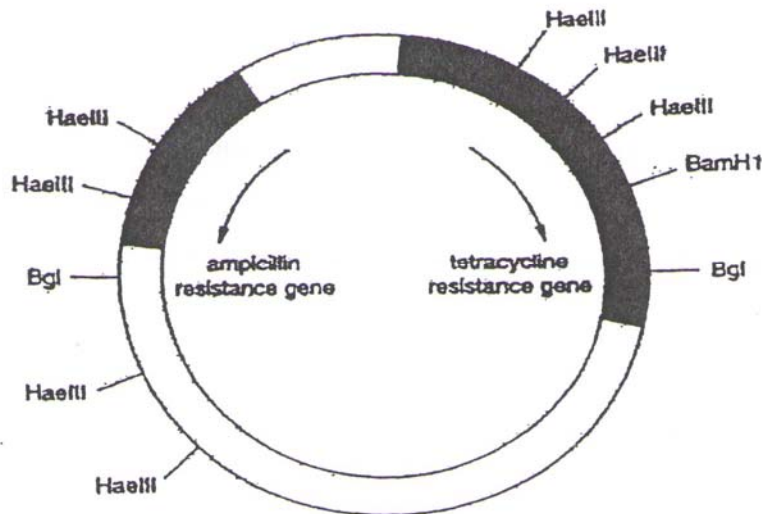


Fig 1.1

i) Describe the function of restriction endonuclease.

[1]

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ii) State which enzyme(s) would be the most suitable for the cleavage of the plasmid and the DNA coding for gene of interest? Explain briefly. [1]

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b) The DNA containing the gene coding for insulin and plasmid with genes coding for resistance against tetracycline, ampicillin and streptomycin were mixed together with the appropriate restriction enzyme. A heat shock treatment was applied to bacteria to allow the plasmids to enter the cells. The bacteria was plated and grown in the following media as shown in Fig. 1.2.

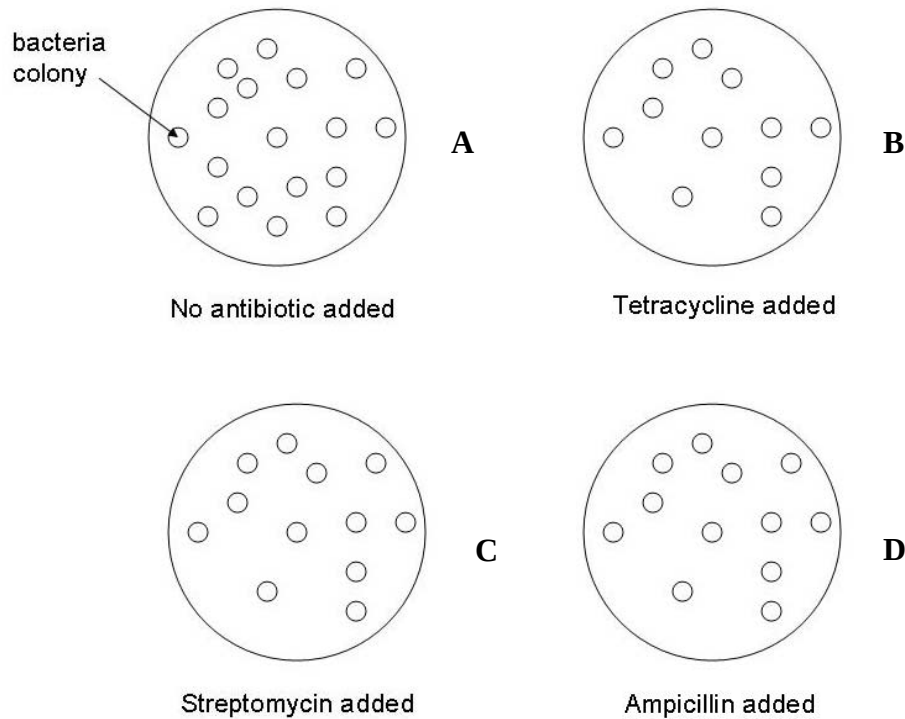


Fig. 1.2

i) How is the gene for human insulin **usually** obtained? Describe the process. [2]

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ii) Explain the advantage of using the process described in part (i). [1]

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iii) Explain why 2 marker genes are needed for identification of transformed cells. [2]

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iv) With reference to Fig. 1.2 above, state which plate contains the gene coding for insulin inserted. [1]

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v) Explain your answer to part (iv). [2]

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vi) Can the insulin obtained directly from the bacteria be used in humans? Why? [2]

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c) Distinguish between a genomic and cDNA library. [3]

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[total: 15]

Question 2

A length of DNA from one of a pair of homologous chromosomes is shown in Fig. 2.1. The target sites of EcoR1 are shown by arrows and the length of DNA between the target sites is given in kilobases (kb).

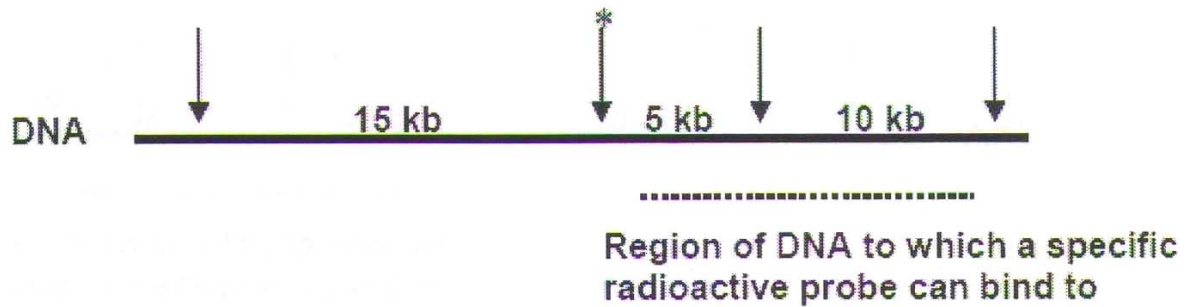


Fig. 2.1

A mutation alters one base of coding sequence at the site marked with an asterisk (*). DNA from two individuals, one who is normal and the other heterozygous for this mutation is cut with EcoR1 and the DNA fragments separated by gel electrophoresis.

a) On Fig 2.2 below,

- i) Draw all the fragments of this length of DNA produced from the pair of homologous chromosomes in each of the individual tested and label the size of the fragments produced. [3]
- ii) Circle the fragments which will bind to the radioactive probe and appear as black bands on an autoradiography. [2]
- iii) Add an arrow to show the direction in which the fragments moved during electrophoresis. [1]

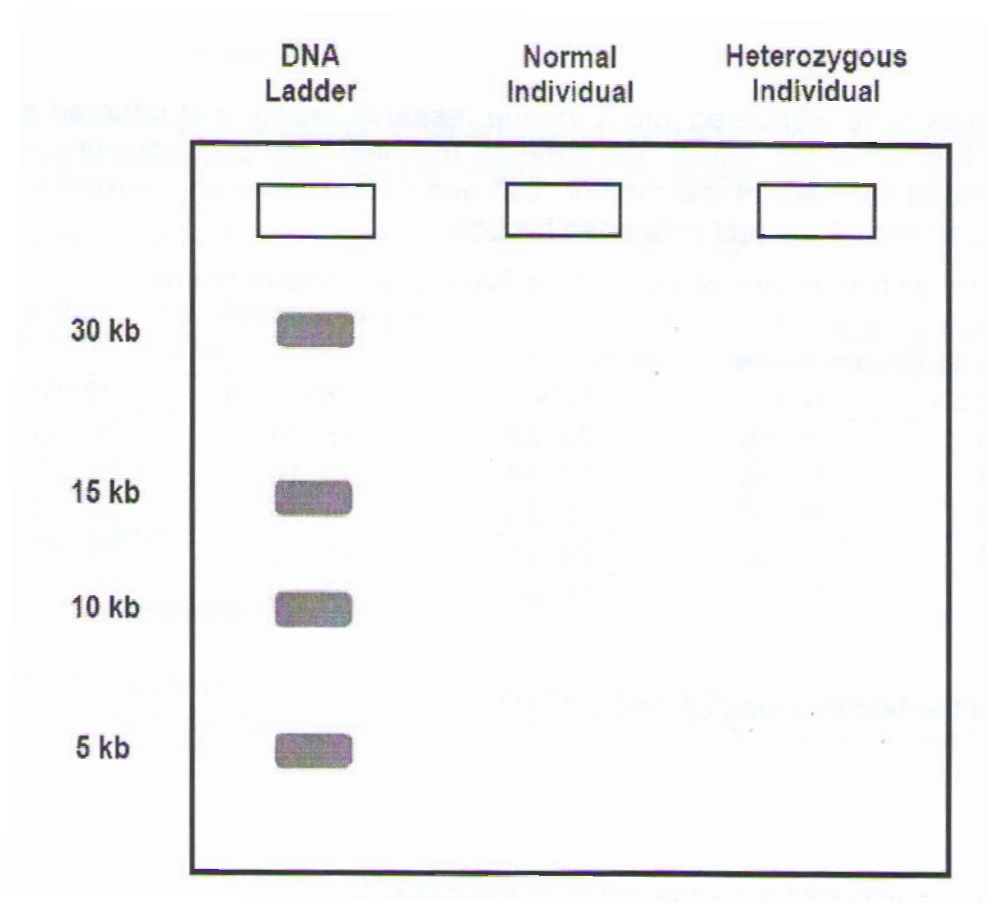


Fig. 2.2

b) State the purpose of DNA ladder.

[1]

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c) Explain how the DNA fragments can be separated by electrophoresis.

[4]

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d) Three male whales have wandered into a marine research facility and fathered a calf. In order to find out which of the males have fathered the new-born calf, scientist extracted DNA from the blood samples of the mother, calf and the three males and carried out a test.

	Locus on Chromosome				
	D3S1358	vWA	FGA	D8S1179	D21S11
Mother	16, 16	15, 16	20, 22	13, 14	28, 30
Calf	16, 18	16, 16	22, 24	13, 14	28, 29
Male 1	16, 18	16, 16	22, 23	14, 15	28, 31
Male 2	17, 18	15, 16	24, 25	11, 12	28, 30
Male 3	15, 18	16, 16	19, 24	12, 13	29, 31

i) State the name of the test that was carried out. [1]

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ii) Which male, if any, may possibly be the father of the calf? [1]

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iii) The type of test performed above can only establish that a male is likely to be the father of a given offspring to a given probability. Explain why it cannot confirm that a given male is the father with absolute certainty. [2]

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[total: 15]

Question 3

Cystic Fibrosis (CF) is one of the most common lethal inherited diseases in humans. Worldwide there are approximately 50,000 sufferers, with an additional 10 million carriers. CF is a result of a single gene mutation and demonstrates autosomal, recessive inheritance.

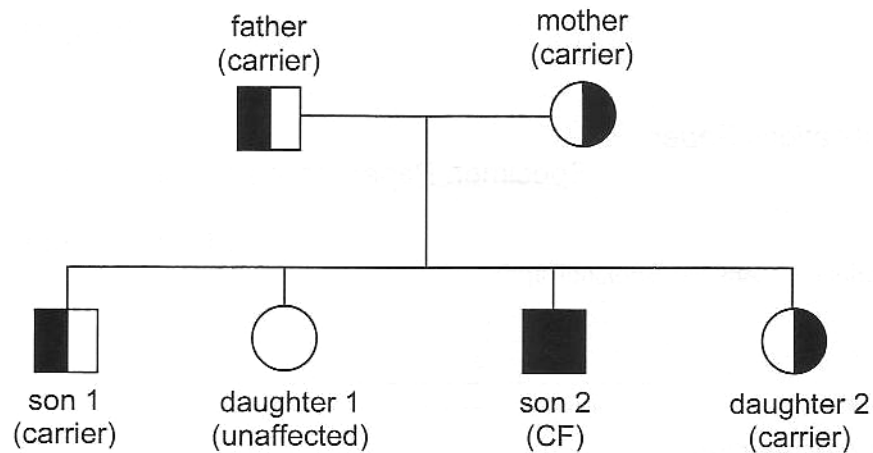


Fig. 3.1

Figure 3.1 shows the inheritance of the disease in a family in which both parents are carriers.

a) Explain how the diagram in Figure 3.1 shows that the disease is

(i) recessive

[2]

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(ii) autosomal

[2]

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The CF gene is found on chromosome 7 and in non-sufferers it gives rise to a protein called Cystic Fibrosis Transmembrane Conductance Regulator (CFTR). The defect which most commonly leads to the disease involves the deletion of three nucleotides from the gene. This is illustrated in Figure 3.2.

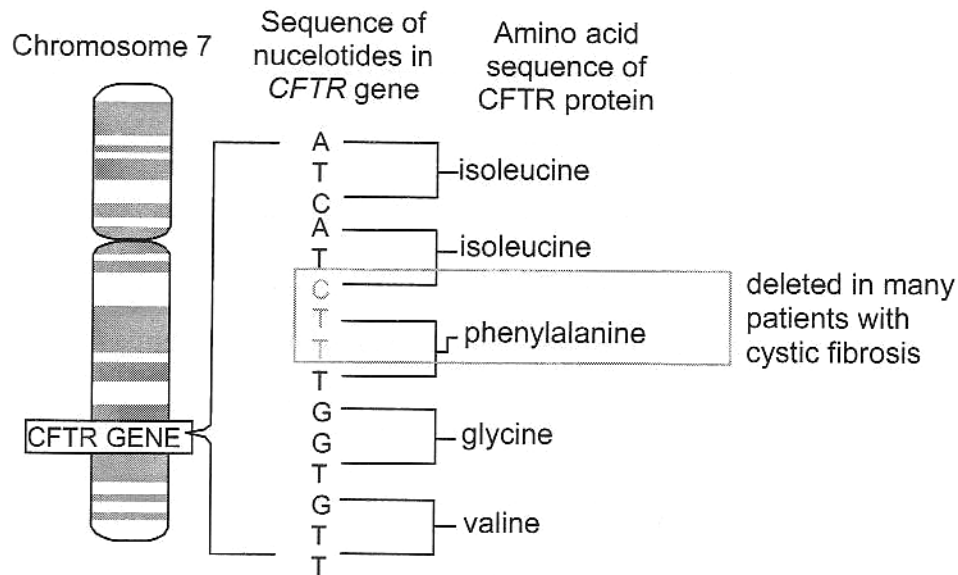


Fig 3.2

- b) With reference to Figure 3.2, explain how this deletion results in the production of a defective protein, leading to cystic fibrosis. [3]

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One type of treatment for cystic fibrosis involves gene therapy, using liposome as a vector. Figure 3.3 below shows the 2 routes possible for delivery of normal gene into a cystic fibrosis patient via liposomes.

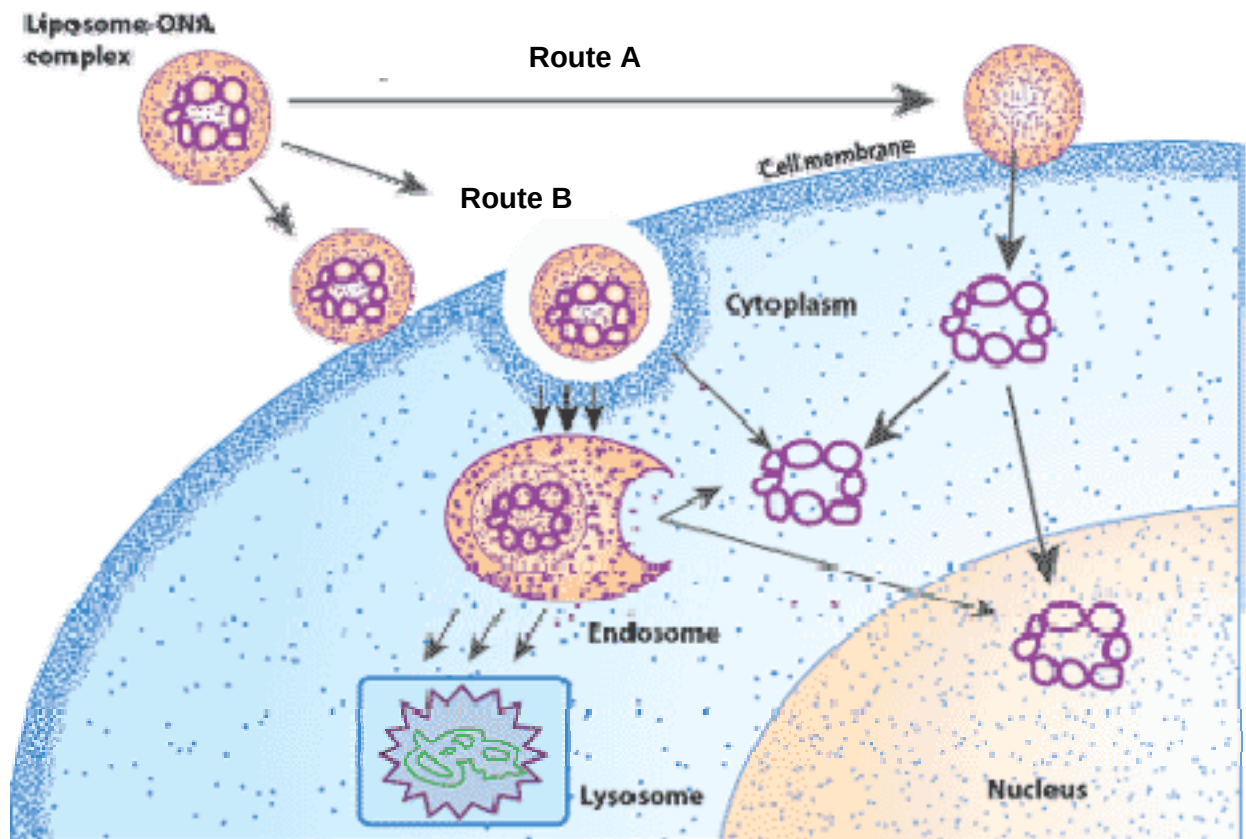


Fig. 3.3

- c) With reference to Fig. 3.3, describe the 2 possible routes of gene delivery into cells using liposomes during gene therapy. [4]

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Besides using liposomes for gene delivery, viral methods have also been used.

d) State 2 different *types* of viral vectors, and describe one advantage and one disadvantage of each. [4]

Viral type :.....

Advantage:
.....

Disadvantage:
.....

Viral type:

Advantage:
.....

Disadvantage:
.....

[15 marks]

Write your answers to this question on the separate answer paper provided.
Your answer should be illustrated by large, clearly labeled diagrams, where appropriate.
Your answer must be in continuous prose, where appropriate.
Your answer should be set out in sections (a), (b) etc., as indicated in the question.

4

- a) Explain the advantages and limitations of polymerase chain reaction (PCR). [6]
- b) Explain the functions of stem cells in a living organism, using appropriate examples to illustrate. [8]
- c) Discuss the advantages and disadvantages of micropropagation/plant tissue culture. [6]